

0.1 Challenge 56: Dark matter detection with Cosmic Explorer

Problem ID: Challenge_56_main

Primary inferred discipline: Gravitation, Cosmology & Astrophysics

Secondary inferred disciplines: High Energy & Nuclear Physics

Python implementation: challenges/56/answer.py

Problem

Problem setup

Consider a scalar field ϕ linearly coupled to the standard model electromagnetic field with coupling strength Λ_γ^{-1} . We can write the interaction term of the Lagrangian as $\mathcal{L}_{\text{int}} = \frac{\phi}{4\Lambda_\gamma} F^{\mu\nu} F_{\mu\nu}$. Assuming the scalar field makes up all the dark matter, it may induce observable effects in laser interferometers. We consider signals of scalar field in the Cosmic Explorer, whose planned strain sensitivity is about $2 \times 10^{-25} \text{ Hz}^{-1/2}$ in the frequency range of $50 \text{ Hz} \leq f \leq 500 \text{ Hz}$. The interferometer arm length of Cosmic Explorer is designed to be 40 km. We assume the Cosmic Explorer uses a similar beamsplitter as LIGO, with thickness 6 cm and index of refraction 3.5.

Main problem

If there is an overdensity of dark matter at the device that is 178 times the local dark matter density (take the local dark matter density to be 0.4 GeV/cm^3 and velocity to be 230 km/s), what is the smallest Λ_γ^{-1} the Cosmic Explorer can probe at 200 Hz with a signal-to-noise ratio of 1 and with an observation time of 1000 s or 0.7 yrs, respectively? Keep at least three significant digits in the final result.

Reviewer's note. The AI-generated solution has been replaced in full by the solution below, which was derived independently before the AI text was read, as the evaluation instructions suggest. The final answer follows the solution. An item-by-item assessment of the challenge statement, of the AI-generated solution and of its numerical script, with the AI text and script reproduced verbatim, follows the final answer in the accompanying document.

Solution

Overview, conventions, and reproducibility

The solution is provided in the following steps:

- Step 1: The dark-matter field amplitude is obtained from the stated density.
- Step 2: The photon coupling turns the field into an apparent oscillation of α .
- Step 3: The oscillating α changes the thickness and the refractive index of the beamsplitter.
- Step 4: Those changes produce a differential optical path in the Michelson interferometer.
- Step 5: The differential path, sampled at the finite light-travel times of a 40 km instrument, is converted to the equivalent strain against which the noise curve is quoted.
- Step 6: The finite coherence of the field fixes how the signal-to-noise ratio (SNR) grows with observation time.
- Step 7: The smallest detectable Λ_γ^{-1} follows.

Throughout, $L = 40$ km is the arm length, $l = 6$ cm and $n = 3.5$ the thickness and refractive index of the beamsplitter, $\sqrt{S_h}$ the strain amplitude spectral density of the detector, T the observation time, and $f = \omega/2\pi = 200$ Hz the frequency probed.

Three conventions are fixed up front so that no ambiguity remains. (i) Λ_γ^{-1} is reported in GeV^{-1} , the unit used in the interferometer literature [1, 2]. (ii) The strain sensitivity $\sqrt{S_h} = 2 \times 10^{-25} \text{ Hz}^{-1/2}$ is a one-sided amplitude spectral density, as all published strain-noise curves are; the SNR of a signal of amplitude h_0 observed for a time T is then $\text{SNR}^2 = 4 \int_0^\infty |\tilde{h}(f)|^2 / S_h(f) \, df = h_0^2 T / S_h$ for a monochromatic signal (derived in Step 6). (iii) 0.7 yr is 0.7 Julian years, $2.2090 \times 10^7 \text{ s}$.¹

Every number below is produced by `challenge56.py`, which lists all problem inputs and all model inputs (with their sources) at its top, uses `astropy` (CODATA 2018) for physical constants, ray-traces the interferometer in 40-digit arithmetic, and prints its internal consistency checks under a `-verbose` flag. The problem's own numbers are used exactly as given, even where they differ slightly from measured literature values (for example $n = 3.5$ versus the measured $n = 3.45$ of silicon at $2 \mu\text{m}$ [3]); measured material data enter only where the problem prompt does not already provide a value explicitly.

Step 1: The dark-matter field

A coherently oscillating scalar field of mass m_ϕ is

$$\phi(t, \mathbf{r}) = \phi_0 \cos(\omega t - \mathbf{k} \cdot \mathbf{r}), \quad \omega = \frac{m_\phi c^2}{\hbar}, \quad \mathbf{k} = \frac{m_\phi \mathbf{v}}{\hbar}, \quad (1)$$

where ϕ_0 is the field amplitude, ω the angular Compton frequency, $\mathbf{k}$ the wave vector, $\mathbf{v}$ the velocity of the dark matter relative to the detector, $\hbar$ the reduced Planck constant, and c the speed of light. At $f = \omega/2\pi = 200$ Hz as asked in this problem, we have $m_\phi c^2 = hf = 8.2713 \times 10^{-13} \text{ eV}$.² For a

¹The choice between a Julian year (365.25 d) and a 365 d year changes the final answer in the fourth significant digit only.

²The observed frequency is $f = (m_\phi c^2 + \frac{1}{2} m_\phi v^2) / \hbar$; the kinetic term shifts m_ϕ by $\frac{1}{2}(v/c)^2 = 3 \times 10^{-7}$, far below three significant digits and ignored.

monochromatic field the energy density $\rho = \frac{1}{2}\dot{\phi}^2 + \frac{1}{2}m_\phi^2\phi^2 = \frac{1}{2}m_\phi^2\phi_0^2$ exactly in natural units, so

$$\phi_0 = \frac{\sqrt{2\rho}}{m_\phi}. \quad (2)$$

With $\rho = 178 \times 0.4 \text{ GeV cm}^{-3} = 71.2 \text{ GeV cm}^{-3} = 5.4707 \times 10^{-40} \text{ GeV}^4$ (converted with $(\hbar c)^3$), Eq. (2) yields $\phi_0 = 39.991 \text{ GeV}$.³⁴

Step 2: The coupling as an oscillating fine-structure constant

Adding the dark-sector interaction $\mathcal{L}_{\text{int}} = \phi F^{\mu\nu} F_{\mu\nu} / (4\Lambda_\gamma)$ to the Maxwell term $-\frac{1}{4}F^{\mu\nu} F_{\mu\nu}$ gives the total photon kinetic term $-\frac{1}{4}(1 - \phi/\Lambda_\gamma)F^{\mu\nu} F_{\mu\nu}$. Because the coefficient of $A_\mu J^\mu$ is the physical charge only when the photon kinetic term has its canonical form $-\frac{1}{4}F^{\mu\nu} F_{\mu\nu}$, we restore that form by the field redefinition $A_\mu \rightarrow A_\mu(1 - \phi/\Lambda_\gamma)^{-1/2}$, which transfers the factor to the charge, $e \rightarrow e(1 - \phi/\Lambda_\gamma)^{-1/2}$, so that $\alpha \rightarrow \alpha/(1 - \phi/\Lambda_\gamma)$ and, to first order in ϕ/Λ_γ [see also e.g., 1, 5],

$$\epsilon(t, \mathbf{r}) \equiv \frac{\delta\alpha}{\alpha}(t, \mathbf{r}) = \frac{\phi(t, \mathbf{r})}{\Lambda_\gamma} = \epsilon_0 \cos(\omega t - \mathbf{k} \cdot \mathbf{r}), \quad \epsilon_0 \equiv \frac{\phi_0}{\Lambda_\gamma} = 39.991 \text{ GeV} \times \Lambda_\gamma^{-1}. \quad (3)$$

Only α varies in Eq. (3). At tree level this is a direct consequence of the problem's Lagrangian, which couples ϕ to only $F^{\mu\nu} F_{\mu\nu}$; the electron mass m_e , which sets atomic sizes together with α (Step 3), is untouched. Beyond tree level, holding m_e fixed is a convention rather than a consequence of the Lagrangian: a varying α feeds into m_e through the electromagnetic self-energy, $\delta m_e/m_e \simeq (3\alpha/2\pi) \ln(\mu/m_e) \delta\alpha/\alpha$ with μ the scale at which the mass parameter is held fixed, a coefficient of order 10^{-2} whose value depends on the renormalization scheme. It is conventional instead to treat Λ_γ and the electron coupling Λ_e as independent parameters and constrain one while setting the other to zero [1, 2]; we adopt the same definition of the photon-only scenario, $\delta m_e = 0$, and carry the induced shift as a third-digit uncertainty alongside the relativistic correction of Step 3. The ultraviolet origin of the coupling (e.g., Higgs portal, dilaton, relaxion) does not enter the calculation; Λ_γ^{-1} is the single parameter being bounded, and nothing gravitational is being probed.⁵

Step 3: Response of the beamsplitter material

Thickness. In nonrelativistic quantum mechanics with Coulomb interactions, every length is proportional to the Bohr radius $a_0 = \hbar/(m_e c \alpha)$. The thickness of a solid, N lattice spacings, therefore scales exactly as $l \propto 1/\alpha$. The linear relation $\delta l/l = -\delta\alpha/\alpha$ is thus exact to first order in $\delta\alpha/\alpha \sim 10^{-22}$ [1, 6]. We write

$$\frac{\delta l}{l} = \kappa_l \frac{\delta\alpha}{\alpha}, \quad \kappa_l = -(1 + K_\alpha), \quad (4)$$

where K_α collects the relativistic correction to the α -scaling of atomic sizes. The single-particle estimate of Stadnik and Flambaum [6, Eq. 12] is $K_\alpha = 2(Z\alpha)^2/[\nu(j + \frac{1}{2})]$, with Z the nuclear charge, ν the effective

³Because an ultralight bosonic dark-matter field such as ϕ is a superposition of many waves with random phases, drawn from the velocity distribution of the halo, its amplitude in any one coherence patch is Rayleigh distributed and Eq. (2) gives the root-mean-square value [4]. “SNR = 1” below is therefore an expectation-value statement; an exclusion limit at a stated confidence level would carry an additional factor of about 3, and up to 10 depending on the statistical treatment, in the short-observation regime [4].

⁴Note that the overdensity factor $178 \simeq 18\pi^2$ is conventionally adopted in the astrophysics and cosmology literature as the virial overdensity of a collapsed halo relative to the cosmic mean density, and a dark-matter overdensity gravitationally bound to the Earth or the Sun would have characteristic velocities, both its internal dispersion and its streaming velocity relative to the detector, far below 230 km s^{-1} [1, Sec. III F]. Whether the problem's 230 km s^{-1} is meant as a dispersion or as a streaming velocity is itself not specified; the consequences are quantified in Step 6. The stated inputs do not describe a self-consistent astrophysical scenario, but they are unambiguous as inputs and are used as given.

⁵The gravitational effect of the overdensity itself is negligible: the oscillating-metric strain of a scalar field is $h \sim G\rho/\omega^2 \approx 5 \times 10^{-36}$ at 200 Hz, nine orders of magnitude below the detection threshold of Step 7.

principal quantum number of the valence electron, and j its total angular momentum. For silicon, $Z = 14$, $(Z\alpha)^2 = 0.0104$, $\nu = \sqrt{\text{Ry}/8.15 \text{ eV}} = 1.29$ with $\text{Ry} = 13.606 \text{ eV}$ the Rydberg energy and 8.15 eV the first ionization energy, and $j = \frac{1}{2}$ or $\frac{3}{2}$ for the $3p$ valence shell, giving $K_\alpha = 0.8\%$ to 1.6% , which is not negligible at the precision requested and which only a many-body calculation of the silicon lattice constant could pin down.⁶ The baseline below uses $\kappa_l = -1$, the implicit assumption of the problem, and the band $K_\alpha \in [0.008, 0.016]$ is carried as a systematic.⁷

Importantly, this calculated response is adiabatic. The fundamental longitudinal mode of a 6 cm silicon plate is $f_0 = v_s/2l \approx 70 \text{ kHz}$, with $v_s \approx 8.4 \text{ km s}^{-1}$ the longitudinal sound speed, so the driven-oscillator correction $[1 - (f/f_0)^2]^{-1}$ of Grote and Stadnik [1, Eq. 7] is $1 + 8 \times 10^{-6}$. The freely suspended plate expands about its center of mass, so each face moves by $\delta l/2$ along the plate normal.⁸

Refractive index. In the same nonrelativistic scaling, all electronic excitation frequencies scale as $\omega_{\text{el}} \propto m_e \alpha^2$ and the static dielectric response of the medium in units of a_0 is α -independent, so the index at a fixed laser frequency ω_L is a function $n = F(\omega_L/\omega_{\text{el}})$ with F independent of α [1, Eqs. 9 to 13]. Hence

$$\frac{\delta n}{n} = \kappa_n \frac{\delta \alpha}{\alpha}, \quad \kappa_n = -2 \frac{\omega_L}{n} \frac{dn}{d\omega_L} = +2 \frac{\lambda_L}{n} \frac{dn}{d\lambda_L}, \quad (5)$$

with λ_L the laser wavelength. The laser frequency itself is locked to the common-mode arm length, which is isolated from size changes by the multiple-pendulum suspensions [1]. The residual common-mode change of the arm length from the mirror faces is $w\epsilon/L \sim 10^{-5}\epsilon$, with w the mirror thickness, and thus negligible.⁹ The index term requires a material and a laser wavelength, neither of which the problem specifies. Among substrate materials used in gravitational-wave detectors only silicon has $n \approx 3.5$ in the near-infrared (3.45 at $2 \mu\text{m}$, Frey et al. 3). The laser wavelength is then fixed by the material: silicon is opaque at the $1 \mu\text{m}$ wavelength of current detectors and of Cosmic Explorer Stage 1, so a silicon beamsplitter can only be used with the $2 \mu\text{m}$ laser of the Stage 2 configuration [8, Table 1]; we therefore evaluate the dispersion at $2 \mu\text{m}$. This is an inference from the given index, not a design fact: the Cosmic Explorer reference design lists a material only for the test masses (silica, or silicon for the $2 \mu\text{m}$ variant) and treats the beamsplitters as fused-silica optics [9, Table 8.1 and Sec. 8.3.2], and the LIGO beamsplitter that the problem calls similar is fused silica [1], with $n \approx 1.45$ at $1 \mu\text{m}$; only its 6 cm thickness matches LIGO's. The measured dispersion of silicon at $2.0 \mu\text{m}$ and 295 K is $dn/d\lambda = -0.0382 \mu\text{m}^{-1}$ with $n = 3.4535$ [3], giving $(\omega/n) dn/d\omega = 0.0221$ and

$$\kappa_n = -0.0442. \quad (6)$$

This term has the same sign as κ_l and contributes 4.4% of the signal after the geometric weights of Step 4.¹⁰

⁶For lighter materials such as silica and sapphire this correction is negligible, as noted by Grote and Stadnik [1].

⁷Finite-nuclear-mass corrections affect only the m_e/m_N dependence and are absent here; QED corrections to the atomic size are $\mathcal{O}(\alpha/\pi)$ without a large logarithm and are below the third digit. The radiative shift of m_e itself (Step 2), which would enter $\delta l/l$ with unit weight, is of order 10^{-2} and depends on the renormalization scheme, that is, on the scale at which the electron mass parameter is held fixed (Step 2); it is set to zero by the photon-only convention and belongs with K_α among the third-digit uncertainties.

⁸Suspension resonances (pendulum and violin modes) respond to forces on the center of mass, not to a size change of the suspended optic [1, 7]; they therefore do not enter the thickness term, and the gradient-force term in Step 5 is treated in the free-mass limit, valid well above the pendulum frequency.

⁹This is the point made by Stadnik and Flambaum [6]: if the laser were locked to a rigid resonator whose length scaled with a_0 , adiabatic changes of α would be a change of units and unobservable. Gravitational-wave detectors avoid this because the laser is locked to the arm length between suspended mirrors, which does not scale with a_0 .

¹⁰For the fused-silica beamsplitters of current detectors the index term is at the 10^{-3} level and is customarily dropped [2]; for silicon it is not. The temperature of the beamsplitter is specified neither by the problem nor, for the beamsplitter, by the Cosmic Explorer design [8]; the room-temperature dispersion is used, and the 123 K value (interpolated from the same data, $\kappa_n = -0.0429$) would change the signal by 0.14%. This is recorded as an input uncertainty in the closing paragraph.

Step 4: Differential optical path of a Michelson interferometer with a thick beamsplitter

Let the arms lie along x and y , the beamsplitter center of mass at the origin, and the plate normal $\hat{\mathbf{N}} = (1, -1)/\sqrt{2}$ (incidence angle $\theta_i = 45^\circ$, fixed by the geometry of the Michelson interferometer). The 50% coating is on the face $\hat{\mathbf{N}} \cdot \mathbf{r} = -l/2$, which faces the laser and the y arm; the opposite face is anti-reflection coated [1, Fig. 1]. Following both beams from the laser to the output port: the x beam is transmitted through the substrate on the way out (one pass), and on return enters the substrate, reflects at the coating from the inside, and exits (two passes); the y beam reflects at the coating on the way out and is transmitted through the substrate once on return. The substrate is therefore crossed three times by the x -arm beam and once by the y -arm beam, and the coating is used as a mirror from opposite sides by the two beams. Inside the plate, the ray is refracted by Snell's law to $\theta_r = \arcsin(\sin \theta_i/n) = 11.66^\circ$ [10, Sec. 1.5.1].

Rather than rely on any published closed form, the script traces both rays with exact vector Snell refraction and reflection, with the plate faces at $\pm l/2$ about the fixed center of mass and fixed reference wavefront planes at the laser, at the two end mirrors, and at the photodetector, and differentiates the round-trip optical path difference $\Delta\mathcal{L} = \mathcal{L}_x - \mathcal{L}_y$ numerically in 40-digit arithmetic.¹¹ The numerical result recovers the closed form to better than 10^{-12} :

$$\delta(\Delta\mathcal{L}) = 2 \left[n \cos \theta_r \delta l + l \frac{\partial(n \cos \theta_r)}{\partial n} \delta n \right], \quad \frac{\partial(n \cos \theta_r)}{\partial n} = \cos \theta_r + \frac{\sin^2 \theta_i}{n^2 \cos \theta_r}, \quad (7)$$

with $n \cos \theta_r = 3.4278$ per encounter with the plate for the thickness term and $l \times 1.0211$ per encounter for the index term. Each encounter contributes equally: on the outbound encounter the x beam gains the plate-insertion path $l(n \cos \theta_r - \cos \theta_i)$ while the y beam loses $l \cos \theta_i$ from the approaching coating; on the return encounter the x beam gains the same insertion path plus $l \cos \theta_i$ from the receding coating while the y beam gains the insertion path. The replacement of vacuum by glass ($-\cos \theta_i$) and the motion of the reflecting face ($+\cos \theta_i$) cancel, leaving $n \cos \theta_r$ per encounter.¹²

Step 5: Finite light-travel time, direction dependence, and equivalent strain

The two encounters with the plate are separated by the round-trip time $2L/c = 267 \mu\text{s}$, which at 200 Hz is a phase $2\omega L/c = 0.335$ rad; the plate is hence not in the same state at the two encounters. Expressing the state of the plate at the first and second encounter as $\epsilon(t - 2L/c)$ and $\epsilon(t)$, the signal at the photodetector is

$$\delta(\Delta\mathcal{L})(t) = \mathcal{A} [\epsilon(t - 2L/c) + \epsilon(t)], \quad \mathcal{A} = n \cos \theta_r \kappa_l l + l \frac{\partial(n \cos \theta_r)}{\partial n} \kappa_n n = -0.21515 \text{ m}, \quad (8)$$

whose amplitude at ω is $2|\mathcal{A}| \cos(\omega L/c) \epsilon_0 = 0.42427 \text{ m} \times \epsilon_0$, a 1.4% reduction from the instantaneous value $2|\mathcal{A}|$.¹³

The strain-noise curve is calibrated against gravitational waves, so the signal must be expressed through the same detector response function [12]. For a Michelson interferometer with one round trip, a strain

¹¹For a plane wave the phase accumulated between two wavefront planes equals the optical path along any ray connecting them [10, theorem of Malus and Dupin, Sec. 3.3.3], so the lateral displacement of the refracted beam does not enter the phase; the result is verified to be independent of the positions of the reference planes to machine precision, the exit rays are exactly antiparallel to the arms, and the calibration $\partial\Delta\mathcal{L}/\partial X_E - \partial\Delta\mathcal{L}/\partial Y_E = 4$ for the end-mirror positions X_E, Y_E is recovered. Reversing which face carries the coating exchanges the roles of the two arms and leaves $|\Delta\mathcal{L}|$ unchanged.

¹²In the no-refraction limit $\theta_r \rightarrow \theta_i$, Eq. (7) reduces to $\sqrt{2}(n - \frac{1}{2})$ per encounter, which is the corrected beamsplitter expression of Vermeulen et al. [2, Eq. 5]; the earlier $\sqrt{2}n$ of Grote and Stadnik [1, Eq. 17] omits the replacement of vacuum by glass. Vermeulen et al. [2] then apply a refraction correction of about 6% for their fused-silica beamsplitter; Eq. (7) gives 6.1% for $n = 1.45$. For the $n = 3.5$ beamsplitter of this problem the refraction correction is 19%, and the original $\sqrt{2}n$ overestimates the thickness coefficient by 44% (Table 1).

¹³The same finite-light-travel-time physics was found to matter for dark-photon searches with LIGO [11]. Within each encounter the events of the two beams are separated by at most 1.4 ns, a phase of 2×10^{-6} , so grouping them as “early” and “late” is exact at the precision required; the script uses the exact event times.

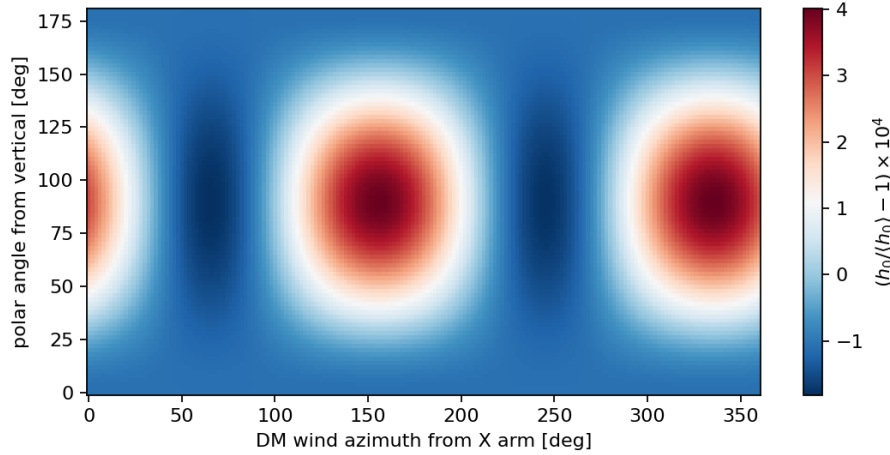


Figure 1: Relative variation of the signal amplitude h_0 with the direction of the dark-matter velocity (azimuth measured from the x arm, polar angle from the vertical), including the end-mirror thickness oscillation and the gradient-force displacement of all optics with their exact sampling times. The variation is at the $\pm 3 \times 10^{-4}$ level and does not affect the third significant digit.

$h(t) = h_0 \cos \omega t$ changes the round-trip optical path of the x arm by $(c/2) \int_{t-2L/c}^t h dt'$ and that of the y arm by the opposite, so $|\Delta \mathcal{L}| = 2Lh_0 \text{sinc}(\omega L/c)$ with $\text{sinc } x = \sin x/x$, the well-known high-frequency correction to the long-wavelength response [12, Eq. 14 at normal incidence]. At 200 Hz and $L = 40$ km, $\omega L/c = 0.1677$ and $\text{sinc} = 0.99532$ (the script verifies this by direct quadrature). The equivalent strain amplitude of the dark-matter signal is therefore

$$h_0 = \frac{2|\mathcal{A}| \cos(\omega L/c)}{2L \text{sinc}(\omega L/c)} \epsilon_0 \equiv \mathcal{H} \Lambda_\gamma^{-1}, \quad \mathcal{H} = \frac{|\mathcal{A}| \cos(\omega L/c)}{L \text{sinc}(\omega L/c)} \phi_0 = 2.1311 \times 10^{-4} \text{ GeV}. \quad (9)$$

Without the two light-travel-time factors $\mathcal{H}$ would be $2.1510 \times 10^{-4} \text{ GeV}$; their net effect is -0.93% , a third-digit effect that the long arms of Cosmic Explorer make unavoidable.

Two further contributions depend on the direction of the dark-matter velocity, which is not known, and are therefore computed over the full sphere of directions (Fig. 1). First, the end mirrors undergo the same thickness oscillation $\delta w/w = \kappa_l \delta \alpha/\alpha$ as the beamsplitter, and their reflecting faces move toward the beamsplitter by $\delta w/2$ with $w = 40$ cm the Stage 2 test-mass thickness [twice the aLIGO value, 1, 8]. The two arms cancel except for the phase difference $\mathbf{k} \cdot (\mathbf{r}_x - \mathbf{r}_y)$ of the field between the end mirrors at $\mathbf{r}_x$ and $\mathbf{r}_y$, which is at most $\sqrt{2} kL$ with $kL = 1.29 \times 10^{-4}$ (reduced de Broglie wavelength 3.1×10^8 m), and which enters in quadrature with the beamsplitter term. Second, the gradient of α exerts a force on every optic through the Coulomb-energy fraction $\kappa_M = 1.7 \times 10^{-3}$ of the silicon atomic mass [1, 5], displacing each free mass by $\delta x = \kappa_M (v/\omega) \epsilon_0 \sin(\omega t - \mathbf{k} \cdot \mathbf{r}) = 0.31 \text{ m} \times \epsilon_0$ along $\hat{\mathbf{v}}$. A common translation of all optics cancels in the differential output, but not exactly, because the optics are sampled at different times ($2L/c$ apart for the beamsplitter's two encounters). This residual reaches 3% of the beamsplitter term for a wind along a plate normal, but it is 90° out of phase and adds in quadrature. The full direction dependence of h_0 is $\pm 3 \times 10^{-4}$ (relative), in the fourth significant digit, and the direction-averaged $\mathcal{H}$ is quoted above.

Step 6: Detection statistics with finite coherence

Monochromatic signal. For $h(t) = h_0 \cos(2\pi f_0 t)$ on $0 < t < T$ the Fourier amplitude near f_0 is $|\tilde{h}(f)|^2 = (h_0^2 T^2/4) \text{sinc}^2[\pi(f - f_0)T]$, whose integral over f is $h_0^2 T/4$. With the one-sided noise convention, $\text{SNR}^2 = 4 \int_0^\infty |\tilde{h}|^2 / S_h df = h_0^2 T / S_h$, giving $\text{SNR} = h_0 \sqrt{T/S_h}$. The script also confirms this analytical expression by a Monte Carlo matched filter on white noise of known one-sided density (0.713 ± 0.018 against the

predicted 0.707; a spurious factor $1/\sqrt{2}$ would have given 0.500). Setting $\text{SNR} = 1$ defines the threshold $h_{\min}^{\text{coh}}(T) = \sqrt{S_h/T}$.

Stochastic signal. The dark-matter field is a superposition of waves with the velocity spread of the halo, so its frequency is spread by $\Delta\omega/\omega \sim (v/c)^2$ and its phase and amplitude wander after a coherence time. Derevianko [13] derived the field autocorrelation for a Maxwellian halo of velocity dispersion ξc seen from a detector moving through it at speed $v_g = \eta \xi c$:

$$\langle \phi(t)\phi(t+\tau) \rangle = \frac{1}{2}\phi_0^2 A(\tau) \cos[\omega'\tau + \psi(\tau)], \quad A(\tau) = \frac{\exp\left[-\frac{\eta^2 x^2}{2(1+x^2)}\right]}{(1+x^2)^{3/4}}, \quad x = \frac{\tau}{\tau_c}, \quad \tau_c = \frac{1}{\xi^2 \omega}, \quad (10)$$

where τ is the time lag, τ_c the coherence time, ω' the Doppler-shifted frequency, and ψ a slowly varying phase. The single velocity given in the problem is identified with the dispersion, $\xi c = 230 \text{ km s}^{-1}$, and the detector speed through the halo is taken equal to it ($\eta = 1$, the fiducial case of Derevianko 13). Then $\tau_c = 1352 \text{ s}$.¹⁴ At $\tau = 1000 \text{ s}$, $A = 0.60$: the field is already substantially decorrelated over the shorter of the two observation times, so the sharp division into a “coherent” and an “incoherent” regime is only an approximation there.

For a Gaussian stochastic signal the optimal statistic is quadratic, and its sensitivity to the coupling is $\sigma_\gamma \propto \left[\sum_p \left(\langle |\tilde{\phi}_p|^2 \rangle / \tilde{\rho}_p \right)^2 \right]^{-1/4}$ [13, Eq. 15], where $\langle |\tilde{\phi}_p|^2 \rangle$ is the expected power of the field in the p -th frequency bin of the discrete Fourier transform and $\tilde{\rho}_p$ the noise power in the same bin (white noise, so independent of p). The expected power spectrum of a record of duration T is the Fourier transform of the correlation function weighted by the triangular window $(1 - |\tau|/T)$, so by Parseval the sum is $\propto \int_0^T (1 - \tau/T)^2 A(\tau)^2 d\tau$, and the phase ψ drops out. Normalizing to the monochromatic threshold, the finite-coherence threshold is

$$h_{\min}(T) = \sqrt{\frac{S_h}{T}} R(T), \quad R(T) = \left[\frac{T/3}{\int_0^T (1 - \tau/T)^2 A(\tau)^2 d\tau} \right]^{1/4}, \quad (11)$$

which tends to 1 for $T \ll \tau_c$ (a fully coherent signal, for which the monochromatic threshold applies) and to $[T/(3I\tau_c)]^{1/4}$ with $I = \int_0^\infty A^2 dx = 0.747$ for $T \gg \tau_c$, reproducing the familiar $\propto (T\tau_c)^{-1/4}$ law with its constant derived rather than assumed.¹⁵ Numerically $R(1000 \text{ s}) = 1.029$ and $R(0.7 \text{ yr}) = 9.241$. The two-regime idealization with $\tau_{\text{coh}} = 2\pi/(m_\phi v^2)$ would give 1 and $(T/\tau_{\text{coh}})^{1/4} = 7.14$ (Fig. 2).

¹⁴Coherence-time conventions differ by factors of order 2π : Grote and Stadnik [1] use $\tau_{\text{coh}} = 2\pi/(m_\phi v^2) = 8495 \text{ s}$ for the same v , Abbott et al. [11] define the linewidth as $\frac{1}{2}(v_0/c)^2 f$. These are labels for the same physics as long as the detection statistic is computed consistently from Eq. (10), which is what is done here. A genuine, not merely notational, ambiguity is whether the problem’s 230 km s^{-1} is the Maxwellian dispersion ξc or the streaming velocity v_g of the detector through the halo. In the direct-detection literature a quoted “dark-matter velocity” of this size is the latter: the Earth’s speed through the halo is $v_E \approx 244 \text{ km s}^{-1}$ in the classic parametrization of Lewin and Smith [14, Eq. 3.6], whose galactic rotation speed v_0 is precisely the problem’s 230 km s^{-1} , and 250 km s^{-1} with current values [15], whereas the dispersion of the standard halo model is $v_0/\sqrt{2} = 168 \text{ km s}^{-1}$ for the recommended circular speed $v_0 = 238 \text{ km s}^{-1}$ [15]. If the problem’s 230 km s^{-1} is interpreted as the streaming velocity (the “streaming reading” below), the dispersion is not supplied by the problem and must be taken from the halo model, giving $\xi c = 168 \text{ km s}^{-1}$, $\eta = 1.37$, $\tau_c = 2525 \text{ s}$, and a 0.7 yr answer lower by 10% (Table 1; the 1000 s answer moves by 1.6%). The dispersion reading, in which the given value is the dispersion and $\eta = 1$ so that both velocities equal the given 230 km s^{-1} , is adopted as primary because it is internally consistent with the problem statement and uses only its own number (it is not a dynamically self-consistent halo: in an equilibrium, non-rotating halo the Jeans equation ties the dispersion to the circular speed, $\sigma_{1D} \approx v_c/\sqrt{2}$, so a streaming speed equal to the dispersion at 230 km s^{-1} would require a halo about 40% hotter than the Milky Way’s rotation curve allows, a net halo rotation of order 70 km s^{-1} , or a local halo out of equilibrium, such as unrelaxed merger debris, for which the Jeans relation does not apply). Note that 230 km s^{-1} is a standard value for the streaming speed and for v_0 , but about 40% above the one-dimensional dispersion of the standard halo model, which is why the streaming reading, the literature-faithful alternative reported alongside, has the longer coherence time. Both readings assume a dispersion-supported halo. A rotation-supported component from dissipative

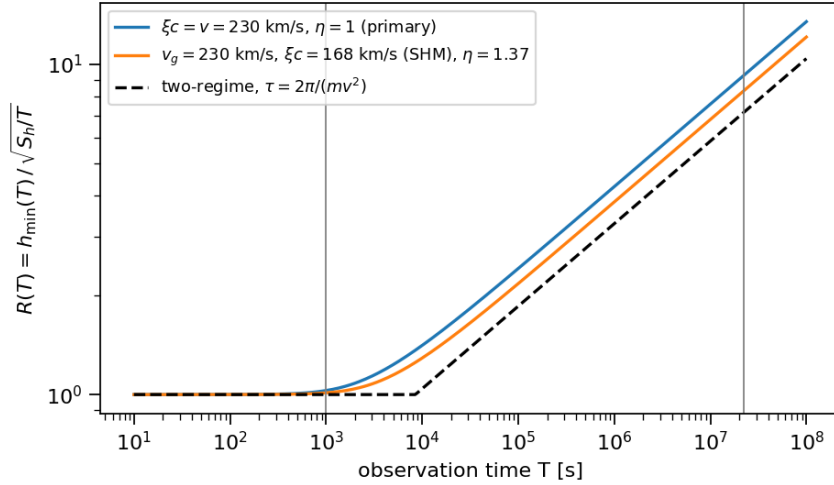


Figure 2: Threshold factor $R(T) = h_{\min}(T)/\sqrt{S_h/T}$ from Eq. (11) for the two readings of the given velocity, compared with the sharp two-regime law implied by the problem. $R = 1$ is the fully coherent limit $T \ll \tau_c$, in which the threshold is the monochromatic $\sqrt{S_h/T}$. The curves depart from it once T approaches τ_c . Vertical lines mark $T = 1000$ s and 0.7 yr.

Step 7: The coupling detection threshold

Combining Eqs. (9) and (11) at SNR = 1,

$$\Lambda_\gamma^{-1}|_{\min} = \frac{h_{\min}(T)}{\mathcal{H}} = \frac{\sqrt{S_h/T} R(T)}{\mathcal{H}}. \quad (12)$$

With $\sqrt{S_h/T} = 6.3246 \times 10^{-27}$ ($T = 1000$ s) and 4.2553×10^{-29} ($T = 0.7$ yr):

$$\Lambda_\gamma^{-1}|_{\min} = 3.05 \times 10^{-23} \text{ GeV}^{-1} (1000 \text{ s}), \quad \Lambda_\gamma^{-1}|_{\min} = 1.85 \times 10^{-24} \text{ GeV}^{-1} (0.7 \text{ yr}). \quad (13)$$

Table 1 shows how the forecast detection thresholds vary under the alternative setups and assumptions discussed above. The largest single effect is the beamsplitter geometry (Step 4), which lowers the thickness coefficient by 31% relative to the uncorrected literature formula; together with the index term and the light-travel-time factors the signal is 28% lower, and the threshold coupling correspondingly higher by a factor 1.39. The finite-coherence statistics (Step 6) are the next largest for the 0.7 yr observation (29% between the two-regime idealization and Eq. (11)).

What the problem idealizes

Three physical features of Cosmic Explorer are outside the problem as posed and are therefore not in the primary answer. (i) *Fabry-Perot arms*. The strain curve of Reitze et al. [8] presupposes arm cavities (its 1.4 to 2 MW circulating power exists only inside them), whereas a beamsplitter signal is not resonantly

dark-matter models would have far smaller velocities in both roles and a correspondingly narrower line.

¹⁵Eq. (11) uses the window-averaged power spectrum as the coherent reference. A search that scans the template frequency finely recovers the on-bin value in the coherent limit, which would raise the incoherent-regime constant by $(3/2)^{1/4} = 1.107$, the factor 3/2 being the ratio of the on-bin to the bin-averaged value of $\sum_p \langle |\tilde{h}_p|^2 \rangle^2$ for a monochromatic signal (bin-averaged $\sum_p \text{sinc}^4 = \pi^{-1} \int \text{sinc}^4 x dx = 2/3$, against 1 on bin). We also note that evaluating Derevianko [13]’s Eq. (15) with his own correlation function, as done here, gives an asymptotic constant smaller than the 2.4 quoted in his Eq. (16), which we could not reconcile with the stated normalization of his line-shape function. Both are $\mathcal{O}(1)$ choices inherent to defining an SNR for a stochastic signal and are absent from the problem statement. The two-regime idealization the problem implies is reported alongside in Table 1.

Table 1: Smallest detectable Λ_γ^{-1} (GeV^{-1}) at $\text{SNR} = 1$ and 200 Hz. “Exact” is Eqs. (7), (9), and (11). The last two rows use the published beamsplitter expressions with the two-regime idealization. Systematics on the first row: relativistic correction to the Bohr scaling, -0.8% to -1.5% ; direction of the dark-matter velocity, $\pm 0.03\%$; beamsplitter temperature, $+0.14\%$.

signal model	time-domain model	$T = 1000 \text{ s}$	$T = 0.7 \text{ yr}$
exact (this work)	Eq. (11), $\xi c = v_g = 230 \text{ km/s}$	3.05×10^{-23}	1.85×10^{-24}
exact	Eq. (11), $v_g = 230$, $\xi c = 168 \text{ km/s}$ (SHM)	3.00×10^{-23}	1.66×10^{-24}
exact	two regimes, $\tau_{\text{coh}} = 2\pi/(m_\phi v^2)$	2.97×10^{-23}	1.43×10^{-24}
Vermeulen et al. [2, Eq. 5], no refraction	two regimes, as above	2.49×10^{-23}	1.19×10^{-24}
Grote and Stadnik [1, Eq. 18]	two regimes, as above	2.13×10^{-23}	1.02×10^{-24}

enhanced by arm cavities. The effective signal is suppressed by $1/N_{\text{eff}}$, with N_{eff} the effective number of round trips in the arm cavities, of order 10^{-2} for a LIGO-like finesse at low frequency and rising linearly with frequency above the cavity bandwidth [1, Eqs. 20 and 21]. The problem gives no finesse or signal-recycling parameters, so it can only intend the response function of a simple Michelson interferometer, which is what was used. In a Fabry-Perot Michelson interferometer the end-mirror term of Step 5, which is resonantly enhanced, would also compete with the beamsplitter term [1]. (ii) *Annual modulation*. Over 0.7 yr the Earth’s orbital velocity ($\pm 30 \text{ km s}^{-1}$) modulates both the line center and, through v^2 , the coherence time by tens of percent [2, 15]. The problem fixes a single v . (iii) *The overdensity scenario* (footnote to Step 1). None of these change the calculation requested; they change what “Cosmic Explorer” would actually measure.

Is the problem well posed?

As a statement it is: every input is specified and the requested quantity is defined. At three significant digits it is underdetermined, for reasons that are physical rather than notational. The beamsplitter model (refraction, index variation, the two-sided use of the coating) moves the answer by tens of percent between defensible published treatments; the relativistic correction to the size scaling of silicon and the scheme-dependent radiative shift of m_e are each 1% effects that no closed-form input can fix; and the meaning of “ $\text{SNR} = 1$ with an observation time” for a partially coherent stochastic signal carries $\mathcal{O}(1)$ constants that the statement does not fix. The statement is also not closed in its inputs: the line shape of a wave-like dark-matter signal depends on two velocities, the dispersion of the halo and the streaming speed of the detector through it, and the problem supplies one number without saying which it is. Every way of filling the second slot is an assumption external to the problem, and the choices span 10% in the 0.7 yr answer (Table 1). The primary answer resolves this by setting both velocities equal to the given value, the only option that adds no information, at the price of a velocity pair that no equilibrium model of the local halo produces (Step 6 footnote). The beamsplitter material and laser wavelength are likewise not given, although the index term (4.4% of the signal) and the relativistic correction (1%) require them at the three significant digits requested, and already at two. They are inferred from $n = 3.5$ (silicon at $2 \mu\text{m}$), an index that matches neither the LIGO beamsplitter the statement calls similar ($n = 1.45$) nor the fused-silica beamsplitters of the Cosmic Explorer design [9]. The beamsplitter temperature (0.14%) is also unspecified. The exact chain above removes every approximation that can be removed and states each remaining choice explicitly. Agreement with a reference answer beyond the second digit would test which conventions the reference adopted, not the physics.

Filled template.py

Listing 1: template.py

```
def answer():
```

```

r"""
Return the values of the smallest  $\Lambda\gamma^{-1}$ .

Inputs
-----

None

Outputs
-----

lambda_inv: list[float]
    Smallest coupling strength  $\Lambda\gamma^{-1}$  the Cosmic Explorer can probe at 200
    Hz
    with an observation time of  $\{1000\text{ s}, 0.7\text{ yrs}\}$  and a signal-to-noise
    of 1.
"""

# ----- FILL IN YOUR RESULTS BELOW -----
# Units:  $\text{GeV}^{-1}$ . Computed by Claude_Notes/challenge56.py (exact beamsplitter
# geometry with Snell refraction, silicon index response, finite light-travel
# time of the 40 km arms, finite-coherence detection statistics).
# The sharp two-regime idealization would give  $[2.97\text{e-}23, 1.43\text{e-}24]$ .
lambda_inv = [3.05e-23, 1.85e-24]
# -----

return lambda_inv

```

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Final answer

$$\Lambda_{\gamma}^{-1}|_{\min} = 3.05 \times 10^{-23} \text{ GeV}^{-1} \quad (T_{\text{obs}} = 1000 \text{ s}), \quad \Lambda_{\gamma}^{-1}|_{\min} = 1.85 \times 10^{-24} \text{ GeV}^{-1} \quad (T_{\text{obs}} = 0.7 \text{ yr})$$

Assessment of the challenge statement and of the AI-generated solution

Scope, provenance, and method

The files assessed are those supplied for Challenge 56: `problem.tex`, the AI-generated `solution.tex`, `final_answer.tex`, `numerics.py`, `answer.py`, `metadata.json` and `agent_log.md`. None of them was modified. The AI solution and its script are reproduced verbatim in Appendices A and B. The solution was originally generated by `codex-cli/gpt-5.6-sol` at reasoning effort “high” on 2026-08-15 and is marked “generated-unreviewed”. The agent had web access through a server-side search tool (21 searches, queries logged, returned pages not) and a Python shell without network access (3 commands, one scratch script that the agent deleted before the run ended). Its citations are bare URLs. Its `numerics.py` runs (Python 3.11) and reproduces the twelve-digit values of its text, $3.01239295648 \times 10^{-23}$ and $1.44734540662 \times 10^{-24} \text{ GeV}^{-1}$, and hence its box.

The AI solution is assessed against the reference solution above. The AI-generated results are all reproduced exactly by adopting the AI’s own model in an independent script, so that every discrepancy below is attributed to a named modeling choice and not to arithmetic.

The challenge statement

The statement is well posed in its core: a single scalar coupling, a given field density, a given interferometer geometry and noise, a given frequency and two observation times determine the threshold coupling by an unambiguous chain (Steps 1 to 7 above). Four inputs that the statement does not fix nevertheless enter the answer beyond the second significant figure.

- (i) The beamsplitter material and the laser wavelength. Only the refractive index $n = 3.5$ is given. It identifies silicon at $2 \mu\text{m}$, the Cosmic Explorer Stage 2 design, but this is an inference. The material enters the index-dispersion response κ_n , which is 4.4% of the signal coefficient (Step 3), and the relativistic correction K_α , about 1%.
- (ii) The meaning of the single velocity 230 km s^{-1} . Read as the one-dimensional velocity dispersion it fixes the coherence time τ_c . If it is instead the circular speed v_0 of the standard halo model, τ_c doubles and the 0.7 yr answer moves by 10% (Step 6 and Table 1).
- (iii) The detection statistic beyond the coherence time. The statement says nothing. The sharp two-regime rule used throughout the literature and the finite-coherence statistic $R(T)$ of Step 6 differ by 3% at 1000 s and 23% at 0.7 yr.
- (iv) The definition of “SNR = 1” (amplitude or power, expectation or exclusion). Both solutions use the amplitude convention $h_0 \sqrt{T/S_h} = 1$. A confidence-level exclusion would add a factor of about 3 [4].

The statement therefore determines the answer to two significant figures at 1000 s and to the first figure with a 30% band at 0.7 yr. It cannot be corrected unambiguously (each missing input has a most plausible value but not a unique one), so `problem.tex` is left untouched and the assumptions are stated in the solution. The overdensity factor 178, the 6 cm thickness and the 40 km arm length are consistent with the Cosmic Explorer design and raise no issue.

The AI solution, item by item

Table 2 lists every modeling element of the AI solution against the reference solution, with a classification along two axes. The first axis separates hidden curriculum (an expert input that the statement does not supply and that a solver cannot derive from it), a genuine error of the solver (the AI), and a deficiency of the challenge statement itself (the question prompt). The second gives the severity: consequential if the boxed answer moves beyond the convention band of the statement, minor if it does not, and simplification

if a known effect was dropped, stated or unstated. Both axes are recorded in the “Classification” column. The “Effect” column gives the quantitative change of the AI answer from that item alone, on which the severity is judged.

Table 2: The AI solution for Challenge 56 against the reference solution. “Effect” is the change of the AI’s threshold Λ_γ^{-1} relative to the reference solution from that item alone (negative means the AI claims a better sensitivity).

Item	AI solution	Reference	Classification	Effect
Field amplitude $\phi_0 = \sqrt{2\rho}/m_\phi$; $m_\phi c^2 = hf$	39.9907 GeV; 8.2713×10^{-22} GeV	39.991 GeV; same	correct, same conventions	none
Beamsplitter coefficient	$nl = 0.2100$ m, from Miller et al. [16], Eq. (3)	$n \cos \theta_r l + l \partial_n (n \cos \theta_r) \kappa_n n = 0.2152$ m (Steps 3 and 4)	simplification, stated only through the citation. The cited formula is a secondary transcription of Grote and Stadnik [1], Eq. (17), whose thickness term is $\sqrt{2}nl$, and predates the correction $\sqrt{2}(n - \frac{1}{2})l$ of Vermeulen et al. [2], Eq. (5). Neither primary formula was cross-checked, although the agent log shows the $\sqrt{2}$ form was searched for. The AI value is within 2% of the exact thickness term $n \cos \theta_r l$ by cancellation, not by derivation: the primary formulas would have given -28% and -16% .	+2.4%
Index-dispersion term κ_n	omitted, not mentioned	-0.0442 from the silicon dispersion at $2 \mu\text{m}$	prompt deficiency (material and wavelength not given); hidden curriculum	+4.4% (in the row above)
Snell refraction, sign structure of the plate	absent	$\theta_r = 11.66^\circ$, per-encounter $n \cos \theta_r$	hidden curriculum; minor	-2.1% (in the row above)
Finite light-travel time of the 40 km arms	absent	$\cos(\omega L/c)/\text{sinc}(\omega L/c) = 0.9906$	hidden curriculum; minor	-0.94%
Relativistic K_α , radiative m_e shift	absent	about 1% each, convention-dependent	convention; minor	about 1%
Coherence time	$\tau_c = 1/[f(v/c)^2] = 8494.85$ s	$\tau_{\text{coh}} = 2\pi/(m_\phi v^2) = 8494.9$ s (same); Derivianko $\tau_c = \tau_{\text{coh}}/2\pi = 1352$ s used inside $R(T)$	consistent; the log shows the 2π ambiguity was recognized	none by itself
Statistic beyond τ_c	sharp two-regime rule, $\sqrt{S_h/T}$ then $\sqrt{S_h/(T\tau_c)}^{1/4}$	finite-coherence $R(T)$: 1.029 at 1000 s, 9.24 at 0.7 yr against 7.14	hidden curriculum and convention; consequential at 0.7 yr	-2.8% (1000 s) -22.7% (0.7 yr)
SNR convention	amplitude, $h_0 \sqrt{T/S_h} = 1$	same	consistent	none

continued on next page

Table 2, continued

Item	AI solution	Reference	Classification	Effect
Constants, units, Julian year	correct (1 cm = same $5.0677 \times 10^{13} \text{ GeV}^{-1}$, $h = 4.135667696 \times 10^{-24} \text{ GeV s}$)		correct	none
Precision	twelve digits in the text, six in the box, for inputs given to one to three	three, with the convention band stated	minor, presentation	none
Assumptions and idealizations	none stated beyond the photon-only coupling (no gravity, uniform field, arm cavities, velocity meaning)	Steps 1, 5, 6 and the two closing paragraphs	documentation	none
Net	3.012×10^{-23} , 1.447×10^{-24}	3.05×10^{-23} , 1.85×10^{-24}		-1.3%, -22%

The AI solution is, in short, the sharp two-regime estimate of the literature with the simplest published signal coefficient. Against the two-regime variant of the reference solution (2.97×10^{-23} and 1.43×10^{-24} , Table 1) it is 1.5% high at both observation times, which is the sum of the beamsplitter-coefficient and light-travel rows of Table 2. Against the reference solution itself it is 1.3% low at 1000 s and 22% low at 0.7 yr, the latter entirely from the statistic. No step is a hallucination: every formula it uses is published, and the one citation that could not be identified from its URL alone, [10.1103/PhysRevD.105.103035](https://arxiv.org/abs/10.1103/PhysRevD.105.103035), is Miller et al. [16], whose Eq. (3) does contain the coefficient nl .

The AI numerical script

`numerics.py` (Appendix B) is under forty lines and mirrors the text line by line with exact CODATA constants, and it asserts agreement with `answer.py` to 5×10^{-6} . It runs and passes. It has no inputs block (every value is a literal inside one function), no sanity check of any intermediate quantity, no variant, and no comment on conventions. Adopting its model in an independent script reproduces its two numbers to ten digits, the residual being its truncated Planck constant. As numerical work it is correct and minimal.

Evidence from the agent log

The log records three points that the delivered text does not. First, the agent searched for the primary form of the coefficient (queries "`delta (L_x-L_y)" "sqrt{2}" beamsplitter`" and "`n l" "Lambda_gamma" beamsplitter`") and noted "Evaluating geometric correction impact" and "Analyzing refractive index and approximation errors", so the $\sqrt{2}$ and the index question were seen and then dropped without a word in the text. Second, it wrote "I'm checking the precise 2π convention now, since it materially affects the 0.7-year number", so the coherence-time ambiguity was recognized, and the standard choice was made silently. Third, its scratch script printed a symbolic threshold `sqrt(2)*L*hthr*mass/(2*ell*n*sqrt(rho))`, which is the nl model, and the script was then deleted, so the intermediate values are recoverable only from the log. The pattern is a solver that identified the right questions and documented none of them.

Verdict

The solution text is within expectation for a fast expert sketch: the chain $\phi_0 \rightarrow h_0 \rightarrow$ coherence regime $\rightarrow$ threshold is right, the numbers are right within the model, the units are clean. It is weak on justification: the central coefficient is taken from a secondary source without cross-check, and none of the four dropped effects (index dispersion, refraction, light travel, K_α) is acknowledged, nor is any assumption stated. The

script is correct and minimal. Holistically the AI answer is right to two significant figures at 1000 s and fails at the third through secondary effects of 1 to 4% each that partly cancel, and it is 22% low at 0.7 yr through a convention that the statement does not fix and that the field itself uses. There is no consequential error. On the three-figure precision the challenge asks for, the statement's own inputs do not determine the third figure, and this should be reflected in the rubric.

Recommended reference values and tolerance for grading

Reference: $\Lambda_\gamma^{-1} = 3.05 \times 10^{-23} \text{ GeV}^{-1}$ at 1000 s and $1.85 \times 10^{-24} \text{ GeV}^{-1}$ at 0.7 yr (finite-coherence statistic).
 Accepted alternative: 2.97×10^{-23} and $1.43 \times 10^{-24} \text{ GeV}^{-1}$ (sharp two-regime rule, the literature convention).
 A tolerance of $\pm 5\%$ around either pair at 1000 s covers the geometry, index and K_α conventions, and $\pm 12\%$ at 0.7 yr covers in addition the reading of the velocity. Answers built on the primary-source coefficients without refraction, $\sqrt{2} n l$ or $\sqrt{2}(n - \frac{1}{2})l$, fall 28% or 16% below the reference solution and should be graded as partially correct. Values off by more than a factor of two indicate a wrong field amplitude, a wrong unit conversion, or a fully coherent 0.7 yr integration (which would give 2.0×10^{-25} , a factor 9 too optimistic).

Appendix A: the AI-generated solution and final answer, verbatim

Solution

Step-by-Step Derivation

The scalar field amplitude and beamsplitter differential-displacement response are [given in the interferometer dark-matter literature](#):

$$\phi(t) = \phi_0 \cos(2\pi ft), \quad \phi_0 = \frac{\sqrt{2\rho_{\text{DM}}}}{m_\phi}$$

in natural units $\hbar = c = 1$, and, when only the photon coupling is nonzero,

$$\delta(L_x - L_y) \simeq \Lambda_\gamma^{-1} nl \phi_0 \cos(2\pi ft).$$

Thus the peak strain amplitude is

$$h_0 = \frac{\delta(L_x - L_y)}{L} = \Lambda_\gamma^{-1} \frac{nl}{L} \phi_0.$$

The finite field coherence time and the sensitivity scalings used below agree with the single-interferometer treatment in [Grote and Stadnik](#): sensitivity scales as $T^{-1/2}$ for $T \lesssim \tau_c$, and as $(T\tau_c)^{-1/4}$ for $T \gtrsim \tau_c$.

1. Dark-matter density and mass

The overdense density is

$$\rho_{\text{DM}} = 178(0.4) = 71.2 \text{ GeV cm}^{-3}.$$

Using

$$1 \text{ cm} = 5.06773071616 \times 10^{13} \text{ GeV}^{-1},$$

we obtain

$$\rho_{\text{DM}} = \frac{71.2}{(5.06773071616 \times 10^{13})^3} = 5.47065597038 \times 10^{-40} \text{ GeV}^4.$$

At $f = 200 \text{ Hz}$,

$$m_\phi c^2 = hf,$$

so in energy units,

$$m_\phi = (4.135667696 \times 10^{-24} \text{ GeV s})(200 \text{ s}^{-1}) = 8.27133539200 \times 10^{-22} \text{ GeV}.$$

Therefore,

$$\phi_0 = \frac{\sqrt{2\rho_{\text{DM}}}}{m_\phi} = \frac{\sqrt{2(5.47065597038 \times 10^{-40})}}{8.27133539200 \times 10^{-22}} = 39.9907057704 \text{ GeV}.$$

2. Strain per unit coupling

For

$$n = 3.5, \quad l = 0.06 \text{ m}, \quad L = 4.00 \times 10^4 \text{ m},$$

the strain amplitude is

$$h_0 = \Lambda_\gamma^{-1} \left(\frac{3.5(0.06)}{4.00 \times 10^4} \right) (39.9907057704 \text{ GeV}),$$

or

$$h_0 = (2.09951205295 \times 10^{-4} \text{ GeV}) \Lambda_\gamma^{-1}.$$

Here Λ_γ^{-1} is expressed in GeV^{-1} .

3. Coherence time

The virialized scalar field has a fractional linewidth of order $(v/c)^2$, giving

$$\tau_c \simeq \frac{1}{f(v/c)^2}.$$

With $v = 230 \text{ km s}^{-1}$,

$$\frac{v}{c} = \frac{230}{299792.458} = 7.67197 \times 10^{-4},$$

and hence

$$\tau_c = \frac{1}{200(230/299792.458)^2} = 8494.85046065 \text{ s}.$$

For a one-sided strain amplitude spectral density

$$\sqrt{S_h} = 2.00 \times 10^{-25} \text{ Hz}^{-1/2},$$

the unit-SNR strain threshold is

$$h_{\min} = \begin{cases} \frac{\sqrt{S_h}}{\sqrt{T_{\text{obs}}}}, & T_{\text{obs}} \leq \tau_c, \\ \frac{\sqrt{S_h}}{(T_{\text{obs}}\tau_c)^{1/4}}, & T_{\text{obs}} > \tau_c. \end{cases}$$

4. Observation for 1000 s

Since

$$1000 \text{ s} < \tau_c,$$

the observation is coherent:

$$h_{\min} = \frac{2.00 \times 10^{-25}}{\sqrt{1000}} = 6.32455532034 \times 10^{-27}.$$

Equating $h_0 = h_{\min}$ gives

$$(\Lambda_{\gamma}^{-1})_{\min} = \frac{6.32455532034 \times 10^{-27}}{2.09951205295 \times 10^{-4} \text{ GeV}} = 3.01239295648 \times 10^{-23} \text{ GeV}^{-1}.$$

5. Observation for 0.7 yr

Using $1 \text{ yr} = 365.25 \text{ days}$,

$$T_{\text{obs}} = 0.7(365.25)(86400) = 2.2090320 \times 10^7 \text{ s}.$$

Because $T_{\text{obs}} \gg \tau_c$, the field phases cannot be added coherently over the full observation:

$$h_{\min} = \frac{2.00 \times 10^{-25}}{[(2.2090320 \times 10^7)(8494.85046065)]^{1/4}} = 3.03871912597 \times 10^{-28}.$$

Consequently,

$$(\Lambda_{\gamma}^{-1})_{\min} = \frac{3.03871912597 \times 10^{-28}}{2.09951205295 \times 10^{-4} \text{ GeV}} = 1.44734540662 \times 10^{-24} \text{ GeV}^{-1}.$$

Final Answer

$$\boxed{\Lambda_{\gamma}^{-1}|_{(1000 \text{ s}, 0.7 \text{ yr})} = (3.01239 \times 10^{-23}, 1.44735 \times 10^{-24})}$$

Appendix B: the AI-generated numerical script, verbatim

Listing 2: numerics.py (AI-generated)

```
#!/usr/bin/env python3
"""Compute Cosmic Explorer scalar-dark-matter sensitivity for Challenge 56."""

from __future__ import annotations

import math
import runpy
from pathlib import Path

def sensitivities() -> tuple[float, float]:
    cm_in_inverse_gev = 5.06773071616e13
    density = 178.0 * 0.4 / cm_in_inverse_gev**3
    mass = 4.135667696e-24 * 200.0
    field_amplitude = math.sqrt(2.0 * density) / mass
    strain_per_coupling = 3.5 * 0.06 / 4.0e4 * field_amplitude

    coherence_time = 1.0 / (200.0 * (230.0 / 299792.458) ** 2)
    noise_asd = 2.0e-25
    short_time = 1000.0
    long_time = 0.7 * 365.25 * 86400.0
    short_threshold = noise_asd / math.sqrt(short_time)
    long_threshold = noise_asd / (long_time * coherence_time) ** 0.25
    return short_threshold / strain_per_coupling, long_threshold / strain_per_coupling

def main() -> None:
    values = sensitivities()
    print(f"1000 s: {values[0]:.12e} GeV^-1")
    print(f"0.7 yr: {values[1]:.12e} GeV^-1")
    expected = runpy.run_path(Path(__file__).with_name("answer.py"))["answer"]()
    if any(not math.isclose(value, target, rel_tol=5.0e-6) for value, target in zip(values,
                                          expected)):
        raise AssertionError((values, expected))

if __name__ == "__main__":
    main()
```

Listing 3: answer.py (AI-generated)

```
def answer():
    r"""
    Return the values of the smallest  $\Lambda_{\gamma^{-1}}$ .

    Inputs
    -----
    None

    Outputs
    -----
    lambda_inv: list[float]
```

Smallest coupling strength $\Lambda\gamma^{-1}$ the Cosmic Explorer can probe at 200 Hz with an observation time of $\{1000\text{ s}, 0.7\text{ yrs}\}$ and a signal-to-noise of 1.

"""

----- FILL IN YOUR RESULTS BELOW -----

lambda_inv = [3.01239e-23, 1.44735e-24]

return lambda_inv

Appendix C: the numerical script of the reference solution

Listing 4: challenge56.py

```
#!/usr/bin/env python
"""
challenge56.py -- Challenge 56: dark matter detection with Cosmic Explorer
=====

Question. A scalar field  $\phi$  with  $L_{\text{int}} = \phi^2 / (4 \Lambda_{\text{gamma}})$  makes up
all the dark matter, with a local overdensity of  $178 \times 0.4 \text{ GeV/cm}^3$  and a
velocity of  $230 \text{ km/s}$ . What is the smallest  $\Lambda_{\text{gamma}}^{-1}$  that Cosmic
Explorer (strain ASD  $2\text{e-}25 \text{ Hz}^{-1/2}$ ,  $40 \text{ km}$  arms,  $6 \text{ cm}$  /  $n = 3.5$  beamsplitter)
can probe at  $200 \text{ Hz}$  with  $\text{SNR} = 1$  for observation times of  $1000 \text{ s}$  and  $0.7 \text{ yr}$ ?

Physics chain (one function per step; equation numbers refer to solution.tex)
  Step 1 density -> field amplitude           $\phi_0 = \sqrt{2 \rho}/m$ 
  Step 2 coupling ->  $\delta \alpha/\alpha$        $= \phi/\Lambda$  (derived in the tex, no code needed)
  Step 3  $\alpha$  -> beamsplitter thickness and refractive index ( $\kappa_l, \kappa_n$ )
  Step 4 exact ray-trace of the 45-degree Michelson with a thick plate ->  $d\text{OPL}/dl, d\text{OPL}/dn, \dots$ 
  Step 5 event timing (finite light-travel time), DM-wind direction, GW calibration ->  $h_0$  per unit  $1/\Lambda$ 
  Step 6 detection statistics for a partially coherent stochastic signal ->  $R(T)$ 
  Step 7  $\Lambda^{-1} = \sqrt{S_h/T} R(T) / H$ 

Reproducibility
  * PROBLEM holds every number given in the problem statement, verbatim.
  * MODEL holds every number the problem does NOT give, each with its source.
  * CONV records the conventions adopted (stated in the tex as well).
  * Physical constants come from astropy.constants (CODATA 2018).
  * --verbose prints all intermediate quantities and the internal consistency checks.
  * results.json and figs/*.png are written in the working directory.

Run: conda run -n pyjax python challenge56.py [--verbose] [--no-fig]
"""
import sys, argparse, json
import numpy as np
import mpmath as mp
import astropy
import astropy.units as u
import astropy.constants as const
from scipy.integrate import quad

mp.mp.dps = 40 # the ray-trace runs in 40-digit arithmetic so that central differences
               # with step  $1\text{e-}12 \text{ m}$  are accurate to  $\sim 1\text{e-}16$  (no symbolic algebra needed)

# =====
# INPUTS
# =====
# --- given in problem.tex (do not edit) -----
PROBLEM = dict(
    f_DM      = 200.0 * u.Hz,          # DM oscillation frequency probed
    rho_local = 0.4 * u.GeV / u.cm**3, # local DM density
    overdensity = 178.0,              # rho_at_device / rho_local
    v_DM      = 230.0 * u.km / u.s,    # the single "velocity" given
    sqrt_Sh   = 2.0e-25 / u.Hz**0.5,  # CE strain ASD, flat over 50-500 Hz (one-sided)
    L_arm     = 40.0 * u.km,           # CE arm length
    l_BS      = 6.0 * u.cm,            # beamsplitter thickness
    n_BS      = 3.5,                   # beamsplitter refractive index (used as given)
    SNR       = 1.0,
    T_obs     = [1000.0 * u.s, 0.7 * u.yr], # astropy's yr is the Julian year (365.25 d)
```

```

)

# --- NOT given in the problem; each with its source -----
MODEL = dict(
    theta_i_deg = 45.0,          # beamsplitter incidence angle: Michelson with perpendicular arms
    lambda_laser = 2.0 * u.um,   # INFERRED: n = 3.5 -> silicon (only substrate with this index) -> opaque
                                # at 1 um
                                # -> 2 um laser of CE Stage 2 (Reitze+2019 Table 1). Not a CE design fact:
                                # the Horizon
                                # Study (Evans+2021, Table 8.1, Sec. 8.3.2) lists silicon only for test
                                # masses and treats
                                # beamsplitters as fused silica; LIGO's beamsplitter is fused silica (n =
                                # 1.45).
    # Frey, Leviton & Madison 2006 (Proc. SPIE 6273), Tables 2 and 3: measured n and dn/dlambda
    # of silicon at 2.0 um. Only the DISPERSION is used (for delta n / n); n itself is the problem's 3.5.
    Si_frey = {295: dict(n=3.45352, dndl=-0.03819), # room temperature: LIGO-like suspended beamsplitter
              (baseline)
              100: dict(n=3.42765, dndl=-0.03664), # bracketing values for the 123 K Stage-2 variant
              150: dict(n=3.43234, dndl=-0.03689)},
    T_BS_baseline = 295,
    # Salzberg & Villa 1957 Sellmeier fit for Si (refractiveindex.info), cross-check of the dispersion
    # only
    Si_sellmeier = dict(B=[10.6684293, 0.0030434748, 1.54133408], C=[0.301516485, 1.13475115, 1104.0]),
    w_ETM = 40.0 * u.cm,        # CE test-mass thickness: aLIGO 20 cm (G&S 2019) x 2 (Reitze+2019
    # Sec. 4.2.1)
    v_s_Si = 8.43 * u.km / u.s, # longitudinal sound speed of Si; used only for the adiabaticity
    # check
    Z_Si = 14,                  # nuclear charge of Si (relativistic correction, Coulomb mass
    # fraction)
    A_Si = 28,                  # mass number of Si
    IE_Si = 8.1517 * u.eV,      # first ionization energy of Si -> effective principal quantum
    # number
    a_C = 0.7 * u.MeV,          # semi-empirical Coulomb coefficient (G&S 2019 Eq. 15; Damour &
    # Donoghue 2010)
    v0_SHM = 238.0 * u.km / u.s, # standard-halo-model circular speed (Baxter+2021); dispersion = v0/
    # sqrt2.
                                # Used only in the "streaming reading" variant, where the problem's
                                # 230 km/s is the detector's speed through the halo (Lewin & Smith
                                # v_E).
)

# --- conventions (also stated in the tex) -----
CONV = dict(
    PSD = "sqrt_Sh is a one-sided ASD; SNR^2 = h0^2 T / S_h for a monochromatic signal (derived, MC-
    checked)",
    strain = "h_equiv = |dOPL_roundtrip(omega)| / |H_GW(omega)|, H_GW = 2 L sinc(omega L/c) (Michelson,
    one round trip)",
    tau_intended = "two sharp regimes with tau_coh = 2 pi/(m v^2) = 1/(f beta^2) (Grote & Stadnik 2019)
    ",
    tau_derevianko = "tau_c = 1/(xi^2 omega) with xi c = v, eta = v_g/(xi c) = 1 (Derevianko 2018 Eq. 4)"
    ,
)

hbar, c, G = const.hbar, const.c, const.G
hbarc = (hbar * c).to(u.GeV * u.cm) # converts GeV/cm^3 to GeV^4
alpha_fs = const.alpha.value

def banner(s):
    print("\n" + "=" * 78 + "\n" + s + "\n" + "=" * 78)

```

```

# =====
# Step 1: field amplitude from the stated density
# =====
def step1_field(P, verbose):
    """phi = phi0 cos(omega t - k.r); rho = m^2 phi0^2 / 2 exactly for a monochromatic field."""
    omega = 2 * np.pi * P["f_DM"]
    m_phi = (hbar * omega).to(u.GeV) # m c^2 = hbar omega (kinetic shift 0.5 beta^2 ignored)
    rho = (P["overdensity"] * P["rho_local"]).to(u.GeV / u.cm**3)
    rho_nat = (rho * hbar**3).to(u.GeV**4) # natural units
    phi0 = (np.sqrt(2 * rho_nat) / m_phi).to(u.GeV) # RMS amplitude (Rayleigh-distributed per coherence patch)
    if verbose:
        banner("Step 1: field amplitude")
        print(f"omega = {omega:.6e}; m_phi c^2 = {m_phi:.6e} = {m_phi.to(u.eV):.6e}")
        print(f"rho = {rho:.6e} = {rho_nat:.6e}; phi0 = {phi0:.6f} (expect ~40 GeV)")
        print(f"check: m^2 phi0^2/2 = {(m_phi**2*phi0**2/2).to(u.GeV**4):.6e} == rho_nat")
        beta = (P['v_DM'] / c).decompose().value
        print(f"Doppler/kinetic correction to f: 0.5 beta^2 = {0.5*beta**2:.2e} (ignored, below 3 s.f.)")
    return dict(omega=omega, m_phi=m_phi, rho=rho, rho_nat=rho_nat, phi0=phi0)

# =====
# Step 3: response of the beamsplitter material to delta alpha / alpha = eps
# delta l / l = kappa_l eps, kappa_l = -(1 + K_alpha)
# Bohr scaling (all lengths prop. to 1/alpha) is exact in nonrelativistic QM;
# K_alpha = 2 (Z alpha)^2 / (nu (j + 1/2)) is the single-particle relativistic
# correction (Stadnik & Flambaum 2015 Eq. 12), carried as a systematic band.
# delta n / n = kappa_n eps, kappa_n = -2 (omega/n)(dn/domega) = +2 (lambda/n)(dn/dlambda)
# because n = F(omega_L / omega_el) with omega_el prop. to m_e alpha^2 and the laser
# frequency held fixed by the arm-length lock (G&S 2019 Eqs. 11-13).
# =====
def si_sellmeier(lam_um, S):
    return np.sqrt(1 + sum(b * lam_um**2 / (lam_um**2 - cc**2) for b, cc in zip(S["B"], S["C"])))

def step3_response(P, M, verbose):
    lam = M["lambda_laser"].to(u.um).value
    fr = M["Si_frey"]
    n295, dnd1295 = fr[295]["n"], fr[295]["dnd1"]
    # 123 K (Stage-2 test-mass temperature) by linear interpolation between the 100 K and 150 K rows
    n123 = fr[100]["n"] + (fr[150]["n"] - fr[100]["n"]) * 23 / 50
    dnd123 = fr[100]["dnd1"] + (fr[150]["dnd1"] - fr[100]["dnd1"]) * 23 / 50
    disp = {295: -(lam / n295) * dnd1295, 123: -(lam / n123) * dnd123} # (omega/n)(dn/domega) > 0 (normal dispersion)
    kappa_n = {T: -2.0 * d for T, d in disp.items()}
    # cross-check of the dispersion with the Salzberg-Villa Sellmeier fit
    h = 1e-4
    S = M["Si_sellmeier"]
    disp_sv = -(lam / si_sellmeier(lam, S)) * (si_sellmeier(lam + h, S) - si_sellmeier(lam - h, S)) / (2 * h)
    # relativistic correction band: valence 3p electron of Si, nu from the ionization energy, j = 1/2 or 3/2
    nu = np.sqrt((const.Ryd * const.h * c).to(u.eV) / M["IE_Si"]).value
    K_alpha = {j: 2 * alpha_fs**2 * M["Z_Si"]**2 / (nu * (j + 0.5)) for j in (0.5, 1.5)}
    kappa_l = -1.0 # baseline (nonrelativistic Bohr scaling)
    # adiabaticity: driven-oscillator factor 1/(1 - (f/f0)^2) for the plate's longitudinal mode (G&S Eq. 7)
    f0_BS = (M["v_s_Si"] / (2 * P["l_BS"])).to(u.Hz)
    nonadiab = 1.0 / (1 - (P["f_DM"] / f0_BS).decompose().value**2)
    # Coulomb-energy fraction of the Si atomic mass: sets the alpha-gradient force on the optics (Step 5)
    m_N = ((const.m_p + const.m_n) / 2 * c**2).to(u.MeV)

```

```

E_C = M["a_C"] * M["Z_Si"]**2 / M["A_Si"]**(1 / 3)
kappa_M = (E_C / (M["A_Si"] * m_N)).decompose().value
if verbose:
    banner("Step 3: solid / index response")
    print(f"Frey+06 Si @2.0um: n(295K)={n295}, dn/dl={dnd1295}/um -> (omega/n)dn/domega = {disp[295]:.5f}; "
          f"123K: n={n123:.5f}, dn/dl={dnd123:.5f} -> {disp[123]:.5f}; Salzberg-Villa: {disp_sv:.5f}"
    )
    print(f"kappa_n(295K) = {kappa_n[295]:+.5f}, kappa_n(123K) = {kappa_n[123]:+.5f}")
    print(f"relativistic band: nu_eff = {nu:.4f}, (Z alpha)^2 = {(M['Z_Si']*alpha_fs)**2:.5f}, "
          f"K_alpha(j=1/2) = {K_alpha[0.5]:.5f}, K_alpha(j=3/2) = {K_alpha[1.5]:.5f} "
          f"-> kappa_l in [{1+K_alpha[0.5]:.4f}, {1+K_alpha[1.5]:.4f}]")
    print(f"BS longitudinal mode f0 = {f0_BS:.1f}; non-adiabatic factor = {nonadiab:.8f}")
    print(f"Si Coulomb mass fraction kappa_M = {kappa_M:.3e}")
return dict(kappa_l=kappa_l, kappa_n=kappa_n, K_alpha=K_alpha, disp=disp, f0_BS=f0_BS,
            nonadiab=nonadiab, kappa_M=kappa_M)

# =====
# Step 4: exact ray-trace of the Michelson with a thick beamsplitter
#
# Layout (2D, plane of the arms). Arms along +x (ETMX at x = X_E) and +y
# (ETMY at y = Y_E); laser enters from -x; output port toward -y. The plate
# has its centre of mass at the origin, normal N = (1,-1)/sqrt2 (45 degrees),
# faces at N.r = s0 +/- l/2; the 50 % coating is on the -l/2 face (facing the
# laser and ETMY), the +l/2 face is anti-reflection coated.
#
# Beam X: transmitted at the coating -> ETMX -> back into the plate, reflects
# at the coating from INSIDE, exits -> output. (3 substrate passes)
# Beam Y: reflected at the coating -> ETMY -> back, transmitted once -> output.
#
# The light meets the plate twice, 2L/c apart, and the plate's state differs
# between the encounters; so the "early" (outbound) and "late" (return)
# encounters get their own (l, n, s0). Phase of a plane wave between two
# wavefront planes = optical path along any ray, so lateral walk-off is
# irrelevant; the reference planes (D, D_p) must and do drop out.
#
# Output: round-trip OPL_X - OPL_Y, the final ray directions, and the optical
# distance of every event from the photodetector (-> event times).
# =====
def _vec(x, y): return mp.matrix([x, y])
def _dot(a, b): return a[0] * b[0] + a[1] * b[1]

def _intersect(r0, d, N, s):
    """Point where the ray r0 + t d meets the plane N.r = s, and the distance t (d is a unit vector)."""
    t = (s - _dot(N, r0)) / _dot(N, d)
    return r0 + t * d, t

def _refract(d, N, eta):
    """Vector Snell law; eta = n_in / n_out. N is flipped to point toward the incident side."""
    if _dot(N, d) > 0:
        N = -N
    c1 = -_dot(N, d)
    c2 = mp.sqrt(1 - eta**2 * (1 - c1**2))
    return eta * d + (eta * c1 - c2) * N

def _reflect(d, N): return d - 2 * _dot(d, N) * N

```



```

def _trace(le, ne, se, ll, nl, sl, XE, YE, D, Dp):
    le, ne, se, ll, nl, sl, XE, YE, D, Dp = [mp.mpf(x) for x in (le, ne, se, ll, nl, sl, XE, YE, D, Dp)]
    N = _vec(1, -1) / mp.sqrt(2)
    ex, ey = _vec(1, 0), _vec(0, 1)
    sAe, sBe = se - le / 2, se + le / 2      # early plate faces: A = coating, B = anti-reflection
    sAl, sBl = sl - ll / 2, sl + ll / 2      # late plate faces
    ev = {}                                   # cumulative optical path at each event

    # ---- beam X ----
    r, d = _vec(-D, 0), ex; o = 0
    r, t = _intersect(r, d, N, sAe); o += t; ev["X_early"] = o      # reach coating (early encounter)
    d = _refract(d, N, 1 / ne)
    r, t = _intersect(r, d, N, sBe); o += ne * t                    # pass 1 through the substrate
    d = _refract(d, N, ne)
    r, t = _intersect(r, d, ex, XE); o += t; ev["X_ETM"] = o        # reach ETMX
    d = _reflect(d, ex)
    r, t = _intersect(r, d, N, sBl); o += t; ev["X_late"] = o        # reach AR face (late encounter)
    d = _refract(d, N, 1 / nl)
    r, t = _intersect(r, d, N, sAl); o += nl * t                    # pass 2, to the coating
    d = _reflect(d, N)                                              # reflect at the coating from inside
    r, t = _intersect(r, d, N, sBl); o += nl * t                    # pass 3, back to the AR face
    d = _refract(d, N, nl)
    r, t = _intersect(r, d, ey, -Dp); o += t                         # to the photodetector plane
    oX, dX = o, d

    # ---- beam Y ----
    r, d = _vec(-D, 0), ex; o = 0
    r, t = _intersect(r, d, N, sAe); o += t; ev["Y_early"] = o      # reach coating (early encounter)
    d = _reflect(d, N)                                              # reflect at the coating from outside
    r, t = _intersect(r, d, ey, YE); o += t; ev["Y_ETM"] = o        # reach ETMY
    d = _reflect(d, ey)
    r, t = _intersect(r, d, N, sAl); o += t; ev["Y_late"] = o        # reach coating (late encounter)
    d = _refract(d, N, 1 / nl)
    r, t = _intersect(r, d, N, sBl); o += nl * t                    # single pass through the substrate
    d = _refract(d, N, nl)
    r, t = _intersect(r, d, ey, -Dp); o += t
    oY, dY = o, d
    # optical distance of each event from the photodetector (negative = before arrival)
    times = {k: float(v - (oX if k[0] == "X" else oY)) for k, v in ev.items()}
    return oX - oY, dX, dY, times

def step4_geometry(P, M, verbose):
    th = np.deg2rad(M["theta_i_deg"])
    assert abs(th - np.pi / 4) < 1e-12, "ray-trace is written for the 45 deg Michelson layout"
    names = ["l_e", "n_e", "s_e", "l_l", "n_l", "s_l", "X_E", "Y_E", "D", "D_p"]
    l0, n0, L0 = P["l_BS"].to(u.m).value, P["n_BS"], P["L_arm"].to(u.m).value
    nom = [l0, n0, 0.0, l0, n0, 0.0, L0, L0, 100.0, 100.0]
    f = lambda a: _trace(*a)[0]
    hstep = mp.mpf("1e-12")

    def deriv(i):
        # central difference of dOPL w.r.t. parameter i
        ap, am = list(nom), list(nom)
        ap[i] = mp.mpf(nom[i]) + hstep; am[i] = mp.mpf(nom[i]) - hstep
        return float((f(ap) - f(am)) / (2 * hstep))

    S = {names[i]: deriv(i) for i in range(8)}      # sensitivities (m of round-trip OPL per m)
    dD, dDp = deriv(8), deriv(9)                  # must vanish
    dOPL_nom, dX, dY, times = _trace(*nom)
    theta_r = np.arcsin(np.sin(th) / n0)
    per_pass = n0 * np.cos(theta_r) - np.cos(th)   # textbook plate-insertion
    path_per_pass

```

```

dnc_dn = np.cos(theta_r) + np.sin(th)**2 / (n0**2 * np.cos(theta_r)) # d(n cos theta_r)/dn
cvac = c.to(u.m / u.s).value
t_ev = {k: v / cvac for k, v in times.items()} # seconds before PD arrival
if verbose:
    banner("Step 4: exact ray-trace sensitivities (round trip, m per m)")
    print("early: d1 %.6f dn %.6f ds0 %.6f | late: d1 %.6f dn %.6f ds0 %.6f | dXE %.6f dYE %.6f"
          % (S["l_e"], S["n_e"], S["s_e"], S["l_l"], S["n_l"], S["s_l"], S["X_E"], S["Y_E"]))
    print(f"totals: d1 {S['l_e']+S['l_l']:.6f}, dn {S['n_e']+S['n_l']:.6f}, ds0 {S['s_e']+S['s_l']:.6f}
")
    print(f"final ray directions X,Y (expect 0,-1,0,-1): {[round(float(x),12) for x in (dX[0],dX[1],dY
[0],dY[1])]}")
    print(f"check: independent of reference planes D, Dp: {dD:.1e}, {dDp:.1e}")
    print(f"check: closed form dOPL/dl = 2 n cos(theta_r) = {2*n0*np.cos(theta_r):.6f}; "
          f"dOPL/dn = 2 l d(n cos theta_r)/dn = {2*l0*dnc_dn:.6f}")
    print(f"check: early = late = n cos(theta_r) = {n0*np.cos(theta_r):.6f} "
          f"(per-pass transmission {per_pass:.6f} + coating-face motion cos(theta_i) {np.cos(th):.6f})
")
    print(f"check: no-Snell limit sqrt2 (n - 1/2) = {np.sqrt(2)*(n0-0.5):.6f} per encounter (Vermeulen
+21 Eq. 5); "
          f"G&S Eq.(17) sqrt2 n = {np.sqrt(2)*n0:.6f}")
    print(f"check: GW calibration dXE - dYE = {S['X_E']-S['Y_E']:.6f} (expect 4); static uniform
translation: "
          f"{S['s_e']+S['s_l'] + (S['X_E']-S['Y_E'])/np.sqrt(2):.1e} (expect 0)")
    print("event times before PD arrival [s]: " + ", ".join(f"{k} {v:.4e}" for k, v in t_ev.items()))
return dict(S=S, theta_r=theta_r, per_pass=per_pass, t_ev=t_ev)

```

```

# =====
# Step 5: complex signal amplitude per unit (1/Lambda)
#
# eps(t, r) = eps0 cos(omega t - k.r) with eps0 = phi0/Lambda. Each optic
# is perturbed at its own event time t_j and position r_j; the response is
# the sum of (sensitivity x perturbation) with phase factor exp(-i(omega t_j + k.r_j)).
# * beamsplitter thickness/index (early and late encounters) [in phase, cos]
# * end-mirror faces move by (w/2) eps toward the BS (mirror breathes about its CoM)
# * centre-of-mass displacements from the alpha-gradient force
# delta x = kappa_M (v/omega) eps0 sin(...) along v-hat [quadrature, sin -> factor -i]
#
# The strain-equivalent amplitude divides by the Michelson GW transfer
# function H_GW = 2 L sinc(omega L/c) (round-trip OPL per unit strain).
# =====
def step5_amplitude(P, M, F, R, Gm, verbose, kappa_l=None, kappa_n=None, timing=True,
                    gw_sinc=True, nth=91, nph=181):
    kappa_l = R["kappa_l"] if kappa_l is None else kappa_l
    kappa_n = R["kappa_n"][M["T_BS_baseline"]] if kappa_n is None else kappa_n
    omega = F["omega"].to(1 / u.s).value; phi0 = F["phi0"].value
    L = P["L_arm"].to(u.m).value; l = P["l_BS"].to(u.m).value; n = P["n_BS"]
    w = M["w_ETM"].to(u.m).value
    v = P["v_DM"].to(u.m / u.s).value
    cvac = c.to(u.m / u.s).value
    k = (F["m_phi"] / (hbar * c) * (P["v_DM"] / c)).to(1 / u.m).value # |k| = m v / hbar
    S, t = Gm["S"], Gm["t_ev"]
    dx0 = R["kappa_M"] * v / omega # displacement amplitude per
    unit eps0
    if not timing:
        t = {key: 0.0 for key in t}

    # grid of DM-velocity directions: v-hat = (sin th cos ph, sin th sin ph, cos th), z vertical
    th = np.linspace(0, np.pi, nth); ph = np.linspace(0, 2 * np.pi, nph)
    TH, PH = np.meshgrid(th, ph, indexing="ij")
    vx, vy = np.sin(TH) * np.cos(PH), np.sin(TH) * np.sin(PH)
    phX, phY = k * L * vx, k * L * vy # k.r at ETMX and ETMY (BS at the origin)

```

```

E = lambda tt, kr=0.0: np.exp(-1j * (omega * tt + kr))      # phase of eps at (t_event, r_event)

A_BS = (kappa_l * l * S["l_e"] + kappa_n * n * S["n_e"]) * E(t["X_early"]) \
      + (kappa_l * l * S["l_l"] + kappa_n * n * S["n_l"]) * E(t["X_late"])
A_ETM = (-kappa_l) * (w / 2) * (S["X_E"] * E(t["X_ETM"], phX) + S["Y_E"] * E(t["Y_ETM"], phY)) #
# face moves by -(delta w)/2 = -kappa_l (w/2) eps toward the BS
A_disp = (-1j) * dx0 * (S["X_E"] * vx * E(t["X_ETM"], phX) + S["Y_E"] * vy * E(t["Y_ETM"], phY)
      + (S["s_e"] * E(t["X_early"]) + S["s_l"] * E(t["X_late"]))) * (vx - vy) / np.
      sqrt(2))
A_tot = A_BS + A_ETM + A_disp                                # m of round-trip OPL per unit eps0
x = omega * L / cvac
H_GW = 2 * L * (np.sinc(x / np.pi) if gw_sinc else 1.0)      # numpy sinc(x) = sin(pi x)/(pi x)
h_unit = np.abs(A_tot) / H_GW * phi0                         # strain per unit (1/Lambda) [GeV]
out = dict(h_unit=h_unit, TH=TH, PH=PH, kL=k * L, x=x, H_GW=H_GW,
      h_BS_only=abs(A_BS) / H_GW * phi0,
      A_BS_abs=abs(A_BS), A_ETM_max=np.abs(A_ETM).max(), A_disp_max=np.abs(A_disp).max(), dx0=dx0
)
if verbose:
    banner("Step 5: signal amplitude per unit 1/Lambda_gamma [GeV^-1]")
    print(f"k L = {k*L:.4e} (reduced de Broglie wavelength {1/k:.3e} m); omega L/c = {x:.5f}; "
          f"sinc = {np.sinc(x/np.pi):.6f}; cos(omega L/c) [early/late] = {np.cos(x):.6f}")
    print(f"|A_BS| = {abs(A_BS):.6f} m/eps0 (no timing would be "
          f"{abs(kappa_l*l*(S['l_e']+S['l_l'])+kappa_n*(S['n_e']+S['n_l'])):.6f}")
    print(f"ETM-size term max {np.abs(A_ETM).max():.3e}, displacement term max {np.abs(A_disp).max()
          :.3e} m/eps0")
    print(f"h per unit 1/Lambda: min {h_unit.min():.6e}, max {h_unit.max():.6e}, "
          f"direction range {(h_unit.max()-h_unit.min())/h_unit.mean():.2e}")
return out

# =====
# Steps 6 and 7: detection statistics and the threshold coupling
#
# Monochromatic: SNR = h0 sqrt(T/S_h) (one-sided PSD; Monte-Carlo checked below).
# Stochastic field (Derevianko 2018 Eqs. 3-4):
# <phi(t) phi(t+tau)> = (phi0^2/2) A(tau) cos(omega' tau + psi),
# A(tau) = exp(-eta^2 x^2 / (2(1+x^2))) (1+x^2)^(-3/4), x = tau/tau_c, tau_c = 1/(xi^2 omega).
# Optimal quadratic estimator (Derevianko Eq. 15): sigma prop. to [sum_p (<|phi_p|^2>/rho_p)^2]^(-1/4).
# The expected power spectrum of a record of duration T is the Fourier transform of
# (1-|tau|/T) x correlation; by Parseval the sum is prop. to int_0^T (1-tau/T)^2 A^2 dtau
# and the phase psi drops out. Normalized to the monochromatic threshold:
# h_min(T) = sqrt(S_h/T) R(T), R(T) = [ (T/3) / int_0^T (1-tau/T)^2 A(tau)^2 dtau ]^(1/4)
# R -> 1 for T << tau_c and R -> (T/(3 I tau_c))^(1/4), I = int A^2 dx, for T >> tau_c.
# =====
def A2(tau, tau_c, eta):
    x2 = (tau / tau_c)**2
    return np.exp(-eta**2 * x2 / (1 + x2)) / (1 + x2)**1.5

def R_of_T(T, tau_c, eta=1.0):
    I, _ = quad(lambda tt: (1 - tt / T)**2 * A2(tt, tau_c, eta), 0, T, limit=500,
        points=[tau_c, 3 * tau_c, 10 * tau_c] if T > 10 * tau_c else None)
    return ((T / 3) / I)**0.25

def step67_limits(P, M, F, h_unit_mean, verbose):
    beta = (P["v_DM"] / c).decompose().value
    omega = F["omega"].to(1 / u.s).value
    Sh = (P["sqrt_Sh"]**2).to(1 / u.Hz).value
    tau_GS = 1 / (P["f_DM"].value * beta**2) # 2 pi/(m v^2): Grote & Stadnik convention
    tau_D = 1 / (beta**2 * omega)           # 1/(xi^2 omega) with xi c = v: Derevianko (
    primary)

```

```

# "streaming reading": the problem's 230 km/s is the detector's speed through the halo (v_g);
# the dispersion is then the SHM value v0/sqrt2 and eta = v_g/(xi c)
xi_str = (M["v0_SHM"] / np.sqrt(2) / c).decompose().value
tau_D_v0 = 1 / (xi_str**2 * omega)
eta_str = beta / xi_str
res = {}
for T in P["T_obs"]:
    Ts = T.to(u.s).value
    # two-regime idealization (sharp switch at tau)
    h_int = P["SNR"] * np.sqrt(Sh / (Ts if Ts < tau_GS else np.sqrt(Ts * tau_GS)))
    h_int_D = P["SNR"] * np.sqrt(Sh / (Ts if Ts < tau_D else np.sqrt(Ts * tau_D)))
    # exact finite-coherence statistics, for the two readings of the given velocity
    R1 = R_of_T(Ts, tau_D, 1.0); R2 = R_of_T(Ts, tau_D_v0, eta_str)
    h_ex = P["SNR"] * np.sqrt(Sh / Ts) * R1
    h_ex_v0 = P["SNR"] * np.sqrt(Sh / Ts) * R2
    res[str(T)] = dict(T_s=Ts, h_intended=h_int, h_intended_tauD=h_int_D, R_D=R1, R_D_v0=R2,
                      h_exact=h_ex, h_exact_v0=h_ex_v0)

    if verbose:
        banner(f"Step 6/7: T = {T} = {Ts:.4e} s")
        print(f"tau_GS = {tau_GS:.1f} s (T/tau = {Ts/tau_GS:.3e}); tau_D = {tau_D:.1f} s (T/tau = {Ts/tau_D:.3e}); "
              f"tau_c(streaming reading, xi c = {xi_str*c.to(u.km/u.s).value:.1f} km/s, eta = {eta_str:.3f}) = {tau_D_v0:.1f} s")
        print(f"h_min intended (two-regime, tau_GS) = {h_int:.4e}; same with tau_D = {h_int_D:.4e}")
        print(f"R(T) Derevianko: xi c = v: {R1:.5f}; streaming reading: {R2:.5f}")
        print(f"h_min exact = {h_ex:.4e}; (streaming reading) {h_ex_v0:.4e}")

if verbose:
    print(f"asymptotic check: R(T>>tau_c) -> (T/(3 I tau_c))^(1/4), I(eta=1) = "
          f"{quad(lambda x: A2(x,1.0,1.0), 0, np.inf)[0]:.4f}")
return dict(tau_GS=tau_GS, tau_D=tau_D, tau_D_v0=tau_D_v0, per_T=res)

# =====
# Independent checks (printed under --verbose)
# =====
def checks(P, F, verbose):
    if not verbose:
        return
    banner("Independent checks")
    # (a) gravity of the overdensity is irrelevant
    rho_si = (F["rho"] / c**2).to(u.kg / u.m**3)
    omega = F["omega"].to(1 / u.s).value
    print(f"rho = {rho_si:.3e}; oscillating-metric strain ~ G rho/omega^2 = "
          f"{(G*rho_si/(omega/u.s)**2).decompose().value:.2e}")
    # (b) SNR convention: matched filter on white noise with a known ONE-SIDED PSD S_h = 2 sigma^2 / f_s
    rng = np.random.default_rng(1)
    fs, T = 2000.0, 200.0
    N = int(fs * T); t = np.arange(N) / fs
    Sh = 1.0; sigma = np.sqrt(Sh * fs / 2); h0 = 0.05
    tmpl = np.cos(2 * np.pi * 200.0 * t)
    snr = [np.sum((h0 * tmpl + rng.normal(0, sigma, N)) * tmpl) / (sigma * np.sqrt(np.sum(tmpl**2)))
           for _ in range(3000)]
    print(f"MC matched-filter SNR = {np.mean(snr):.3f} +- {np.std(snr)/np.sqrt(len(snr)):.3f}; "
          f"h0 sqrt(T/S_h) = {h0*np.sqrt(T/Sh):.3f}; a spurious 1/sqrt2 would give {h0*np.sqrt(T/(2*Sh)):.3f}")
    # (c) Michelson GW transfer function: dOPL_X - dOPL_Y = c int_{t-2L/c}^{t} h dt' for h = cos(omega t)
    L = P["L_arm"].to(u.m).value; cv = c.to(u.m / u.s).value
    tau = 2 * L / cv
    amp = max(abs(cv * quad(lambda x: np.cos(omega * x), t0 - tau, t0)[0])
              for t0 in np.linspace(0, 2 * np.pi / omega, 400))
    print(f"GW response: numeric amplitude / (2L) = {amp/(2*L):.6f}; sinc(omega L/c) = {np.sinc(omega*L/c/np.pi):.6f}")

```

```
# =====
def main():
    ap = argparse.ArgumentParser()
    ap.add_argument("--verbose", action="store_true")
    ap.add_argument("--no-fig", action="store_true")
    a = ap.parse_args(); V = a.verbose
    if V:
        banner("Environment")
        print(f"python {sys.version.split()[0]}, numpy {np.__version__}, mpmath {mp.__version__}, "
              f"astropy {astropy.__version__} ({const.hbar.reference})")
        print("PROBLEM:", {k: (str(v) if not isinstance(v, list) else [str(x) for x in v]) for k, v in
        PROBLEM.items()})
        print("MODEL  :", {k: str(v) for k, v in MODEL.items()})
        print("CONV   :", CONV)

    F = step1_field(PROBLEM, V)
    R = step3_response(PROBLEM, MODEL, V)
    Gm = step4_geometry(PROBLEM, MODEL, V)
    A = step5_amplitude(PROBLEM, MODEL, F, R, Gm, V)
    TD = step67_limits(PROBLEM, MODEL, F, A["h_unit"].mean(), V)
    checks(PROBLEM, F, V)

    # ---- variants of the signal amplitude H (strain per unit 1/Lambda) ----
    l = PROBLEM["l_BS"].to(u.m).value; n = PROBLEM["n_BS"]; L = PROBLEM["L_arm"].to(u.m).value
    phi0 = F["phi0"].value
    run = lambda **kw: step5_amplitude(PROBLEM, MODEL, F, R, Gm, False, **kw)["h_unit"].mean()
    hv = {
        "exact (baseline: Frey 295K, kappa_l=-1, timing, sinc)": A["h_unit"].mean(),
        " direction min / max": (A["h_unit"].min(), A["h_unit"].max()),
        " no timing, no sinc": run(timing=False, gw_sinc=False),
        " timing only (no sinc)": run(gw_sinc=False),
        " delta n dropped": run(kappa_n=0.0),
        " BS at 123 K": run(kappa_n=R["kappa_n"][123]),
        " relativistic kappa_l (j=1/2)": run(kappa_l=-(1 + R["K_alpha"][0.5])),
        " relativistic kappa_l (j=3/2)": run(kappa_l=-(1 + R["K_alpha"][1.5])),
        "G&S Eq.(18): sqrt2 n l / L, no dn": np.sqrt(2) * n * l / L * phi0,
        "Vermeulen Eq.(5) no Snell: sqrt2 (n-1/2) l / L": np.sqrt(2) * (n - 0.5) * l / L * phi0,
        "closed form n cos(theta_r) l / L, no dn, no timing": n * np.cos(Gm["theta_r"]) * l / L * phi0,
    }
    banner("Signal amplitude h per unit Lambda^-1 [GeV^-1]: variants")
    for k, v in hv.items():
        print(f"{k:60s} " + ("%.6e / %.6e" % v if isinstance(v, tuple) else f"{v:.6e}"))

    # ---- result table: signal model x time-domain model ----
    banner("RESULT: smallest Lambda_gamma^-1 [GeV^-1] at SNR = 1, f = 200 Hz")
    hdr = f"{'signal model':42s} {'time model':34s} {'1000 s':>12s} {'0.7 yr':>12s}"
    print(hdr); print("-" * len(hdr))
    rows = []
    sig_models = [("exact ray-trace (baseline)", A["h_unit"].mean()),
                  ("G&S Eq.(18) (likely intended)", hv["G&S Eq.(18): sqrt2 n l / L, no dn"]),
                  ("Vermeulen Eq.(5), no Snell", hv["Vermeulen Eq.(5) no Snell: sqrt2 (n-1/2) l / L"])]
    time_models = [("two-regime, tau=2pi/(m v^2)", "h_intended"), ("two-regime, tau=1/(m v^2)", "
    h_intended_tauD"),
                  ("Derevianko exact, xi c = v", "h_exact"), ("Derevianko exact, streaming + SHM", "
    h_exact_v0")]
    keys = list(TD["per_T"].keys())
    for sn, hu in sig_models:
        for tn, tk in time_models:
            vals = [TD["per_T"][kk][tk] / hu for kk in keys]
            rows.append(dict(signal=sn, time=tn, lam_1000s=vals[0], lam_07yr=vals[1]))
```

```

        print(f"{sn:42s} {tn:34s} {vals[0]:12.4e} {vals[1]:12.4e}")
    print(f"\ndirection range (baseline signal, Derevianko exact): 1000 s "
          f"{TD['per_T'][keys[0]]['h_exact']/A['h_unit'].max():.4e} .. {TD['per_T'][keys[0]]['h_exact']/A['h_unit'].min():.4e}")
    print(f"tau_GS = {TD['tau_GS']:.1f} s, tau_D = {TD['tau_D']:.1f} s; R(1000 s) = {TD['per_T'][keys[0]]['R_D']:.4f}, "
          f"R(0.7 yr) = {TD['per_T'][keys[1]]['R_D']:.4f}")

    json.dump(dict(h_unit=hv, limits=rows, tau=dict(GS=TD["tau_GS"], D=TD["tau_D"], D_v0=TD["tau_D_v0"]),
                  R={k: (TD["per_T"][k]["R_D"], TD["per_T"][k]["R_D_v0"]) for k in keys},
                  K_alpha=R["K_alpha"], kappa_n=R["kappa_n"], sens=Gm["S"], t_ev=Gm["t_ev"]),
              open("results.json", "w"), indent=1, default=lambda o: float(o) if np.isscalar(o) else str(o))
    print("wrote results.json")

    if not a.no_fig:
        import matplotlib; matplotlib.use("Agg")
        import matplotlib.pyplot as plt, os
        os.makedirs("figs", exist_ok=True)
        # (1) direction dependence of the signal amplitude
        rel = (A["h_unit"] / A["h_unit"].mean() - 1) * 1e4
        fig, ax = plt.subplots(figsize=(7, 3.6))
        im = ax.pcolormesh(np.rad2deg(A["PH"]), np.rad2deg(A["TH"]), rel, shading="auto", cmap="RdBu_r")
        ax.set_xlabel("DM wind azimuth from X arm [deg]"); ax.set_ylabel("polar angle from vertical [deg]")
    )
    fig.colorbar(im, ax=ax).set_label(r"${h_0/\angle h_0\angle - 1}\times 10^{-4}$")
    fig.tight_layout(); fig.savefig("figs/direction_dependence.png", dpi=160)
    # (2) threshold factor R(T) versus the two-regime idealization
    Ts = np.logspace(1, 8, 300)
    fig, ax = plt.subplots(figsize=(6, 3.6))
    xi_str_kms = MODEL["v0_SHM"].to(u.km / u.s).value / np.sqrt(2)
    for tau, eta, lab in [(TD["tau_D"], 1.0, r"${\xi} c = v = 230$ km/s, ${\eta}=1$ (primary)",
                          (TD["tau_D_v0"], PROBLEM["v_DM"].to(u.km / u.s).value / xi_str_kms,
                           r"${v}_g = 230$ km/s, ${\xi} c = %.0f$ km/s (SHM), ${\eta}=%.2f$" % (xi_str_kms,
230 / xi_str_kms))]:
        ax.loglog(Ts, [R_of_T(T, tau, eta) for T in Ts], label=lab)
    ax.loglog(Ts, np.where(Ts < TD["tau_GS"], 1, (Ts / TD["tau_GS"])**0.25), "k--",
              label=r"two-regime, ${\tau}=2\pi/(mv^2)$")
    for T in PROBLEM["T_obs"]:
        ax.axvline(T.to(u.s).value, color="grey", lw=0.8)
    ax.set_xlabel("observation time T [s]"); ax.set_ylabel(r"${R}(T) = h_{\min}(T)\backslash,\backslash\sqrt{S_h/T}$")
    ax.tick_params(which="major", length=3.5 * 1.5) # 1.5 x matplotlib defaults (3.5 / 2.0 pt)
    ax.tick_params(which="minor", length=2.0 * 1.5)
    ax.legend(fontsize=8)
    fig.tight_layout(); fig.savefig("figs/R_of_T.png", dpi=160)
    print("figures: figs/direction_dependence.png, figs/R_of_T.png")

if __name__ == "__main__":
    main()

```